

Share Issuance Restraint and the Cross Section of Stock Returns

I. M. Harking

October 20, 2025

Abstract

This paper studies the asset pricing implications of Share Issuance Restraint (SIR), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on SIR achieves an annualized gross (net) Sharpe ratio of 0.51 (0.45), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 28 (25) bps/month with a t-statistic of 3.60 (3.27), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (5 year), Growth in book equity, Share issuance (1 year), Momentum and LT Reversal, Long-term EPS forecast, Analyst Value) is 20 bps/month with a t-statistic of 2.32.

1 Introduction

Asset prices reflect investors' marginal utility of consumption across states of nature, with expected returns compensating for systematic exposure to consumption risk. While the consumption-based asset pricing framework provides a fundamental theory for understanding cross-sectional return variation, empirical tests often struggle to identify firm characteristics that cleanly capture differences in consumption risk exposure. We examine share issuance restraint (SIR) as a signal that predicts stock returns through its connection to firms' exposure to aggregate consumption risk. The tendency of firms to limit equity issuance reveals information about their sensitivity to macroeconomic conditions and their ability to buffer consumption shocks, particularly during economic downturns when marginal utility is high.

Share issuance restraint reflects firms' exposure to systematic consumption risk through multiple channels that affect the intertemporal marginal rate of substitution. Firms that exhibit greater restraint in issuing equity demonstrate lower flexibility in adjusting their capital structure during periods of high marginal utility, making them more sensitive to consumption growth shocks ([Campbell and Cochrane, 1999](#)). This inflexibility becomes particularly pronounced during bad economic states when external financing frictions intensify and the representative agent's risk aversion increases through habit formation mechanisms ([Constantinides, 1990](#)). The consumption-based interpretation suggests that firms with high SIR carry greater systematic risk because their cash flows covary more strongly with aggregate consumption during disaster states ([Barro, 2006](#)).

The state-dependent nature of share issuance decisions links directly to long-run consumption risks that drive asset price variation ([Bansal and Yaron, 2004](#)). During good times, when consumption growth is high and marginal utility is low, firms can more easily access equity markets without significant dilution costs. However, firms exhibiting issuance restraint maintain this behavior even in favorable conditions,

suggesting persistent exposure to consumption risk factors that manifest across the business cycle. This persistent exposure translates into higher required returns as investors demand compensation for bearing undiversifiable consumption risk (Breedén and Litzenberger, 1978).

The connection between SIR and disaster risk sensitivity provides an additional consumption-based channel for return predictability. Firms that restrain from issuing shares demonstrate characteristics associated with higher tail risk exposure during consumption disasters (Gabaix, 2012). These firms typically have less financial flexibility to smooth consumption shocks, making their equity claims more sensitive to rare disaster events that dramatically increase marginal utility. The inability to adjust financing during extreme states amplifies the covariance between firm cash flows and the stochastic discount factor, particularly when consumption drops sharply and marginal utility spikes (Rietz, 1988).

The share issuance restraint strategy delivers economically and statistically significant returns that persist after controlling for standard risk factors. The value-weighted long/short SIR portfolio generates an average monthly return of 30 basis points with a t-statistic of 4.01, translating to an annualized Sharpe ratio of 0.51. The strategy’s alpha remains robust across factor models, with the six-factor alpha equaling 28 basis points per month (t-statistic = 3.60), confirming that existing factors do not fully capture the consumption risk exposure embedded in share issuance restraint.

The economic significance of the SIR premium manifests consistently across different portfolio construction methodologies and size groups. Among the largest stocks—those above the 80th NYSE percentile—the strategy earns 21 basis points per month (t-statistic = 2.36), demonstrating that the effect is not confined to small, illiquid securities where consumption risk might be confounded with market microstructure effects. After accounting for transaction costs using high-frequency effective

spreads, the net return equals 26 basis points per month (t-statistic = 3.49) with a net Sharpe ratio of 0.45, indicating that the consumption risk premium associated with SIR is realizable by investors.

Controlling for closely related anomalies and the broader factor zoo reveals that SIR captures unique variation in consumption risk exposure. When we simultaneously control for the six most closely related strategies and the six Fama-French factors, the SIR strategy maintains an alpha of 20 basis points per month (t-statistic = 2.32). The strategy’s gross monthly alpha relative to the entire factor zoo places it in the top 10% of documented anomalies, while its net performance after transaction costs ranks in the top 3%, confirming that share issuance restraint identifies a distinct and economically important source of consumption-based systematic risk.

This paper advances the consumption-based asset pricing literature by identifying share issuance restraint as a powerful predictor of cross-sectional returns that directly captures firms’ exposure to consumption risk. Unlike (Pontiff and Woodgate, 2008) who focus on the mispricing implications of share issuance, we demonstrate that issuance restraint reflects rational compensation for bearing systematic consumption risk, particularly during high marginal utility states. Our findings complement (Daniel and Titman, 2006) by showing that financing decisions reveal information about disaster sensitivity and long-run consumption risks rather than purely behavioral biases. The consumption-based interpretation provides a unified framework that explains both the magnitude and persistence of the SIR premium across different market conditions.

Methodologically, we contribute by demonstrating that share issuance restraint survives the comprehensive testing protocol of recent studies questioning the robustness of asset pricing anomalies. Following (McLean and Pontiff, 2016) and (Hou et al., 2020), we show that the SIR effect persists out-of-sample, across different portfolio construction methods, and after accounting for data-snooping concerns. Our results

withstand the stringent tests proposed by (Harvey et al., 2016) for evaluating new factors, with t-statistics exceeding conventional thresholds even after adjusting for multiple testing. The ability of SIR to generate significant alphas relative to existing consumption-based factors suggests that current models incompletely capture the full spectrum of consumption risk exposure.

The broader implications extend to understanding how corporate financing decisions encode information about systematic risk exposure in a consumption-based framework. Our evidence suggests that seemingly puzzling patterns in equity issuance reflect rational responses to consumption risk rather than market inefficiencies, supporting the view that financial markets efficiently price consumption risk even in complex corporate actions (Cochrane et al., 2008). The persistence of the SIR premium after controlling for momentum, value, and profitability factors indicates that consumption-based explanations provide insights beyond production-based or behavioral models, reinforcing the fundamental role of consumption risk in determining equilibrium asset prices (Cochrane, 2005).

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the requisite variables, spanning several decades of financial reporting. We focus on annual observations to capture year-over-year changes in equity issuance activity. Share Issuance Restraint signal relies on two key variables from COMPUSTAT. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet. This item reflects the book value of common equity at par, which changes when firms issue new shares, repurchase existing shares, or retire treasury stock. The market eq-

uity variable ($me_{atadate}$) represents the market value of common equity, calculated as the product of share count and end date. This variable provides a market-based scaling factor that normalizes the change in common stock over-year change in common stock scaled by lagged market equity : $(CSTK(t) - CSTK(t-1)) / me_{atadate}(t-1)$. This measure captures the relative change in the book value of common shares outstanding. The adjusted measure of equity issuance activity that facilitates cross-sectional comparisons. We calculate the end values and require non-missing observations for both $CSTK$ and $me_{atadate}$ in consecutive years. Finally, positive lagged market equity are excluded from the analysis to avoid undefined or economically meaningless values.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the SIR signal. Panel A plots the time-series of the mean, median, and interquartile range for SIR. On average, the cross-sectional mean (median) SIR is -0.01 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input SIR data. The signal's interquartile range spans -0.01 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the SIR signal for the CRSP universe. On average, the SIR signal is available for 6.35% of CRSP names, which on average make up 7.68% of total market capitalization.

4 Does SIR predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on SIR using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high SIR portfolio and sells the low SIR portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015)

five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short SIR strategy earns an average return of 0.30% per month with a t-statistic of 4.01. The annualized Sharpe ratio of the strategy is 0.51. The alphas range from 0.28% to 0.31% per month and have t-statistics exceeding 3.60 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.18, with a t-statistic of 3.28 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 617 stocks and an average market capitalization of at least \$1,465 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals

25 bps/month with a t-statistics of 3.41. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-one exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 21-27bps/month. The lowest return, (21 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.92. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the SIR trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-five cases.

Table 3 provides direct tests for the role size plays in the SIR strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and SIR, as well as average returns and alphas for long/short trading SIR strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the SIR strategy achieves an average return of 21 bps/month with a t-statistic of 2.36. Among these large cap stocks, the alphas for the SIR strategy relative to the five most common factor models range from 17 to 22 bps/month with t-statistics between 1.90 and 2.44.

5 How does SIR perform relative to the zoo?

Figure 2 puts the performance of SIR in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the SIR strategy falls in the distribution. The SIR strategy’s gross (net) Sharpe ratio of 0.51 (0.45) is greater than 90% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the SIR strategy (red line).² Ignoring trading costs, a \$1 invested in the SIR strategy would have yielded \$6.86 which ranks the SIR strategy in the top 3% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the SIR strategy would have yielded \$4.91 which ranks the SIR strategy in the top 2% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the SIR relative to those. Panel A shows that the SIR strategy gross alphas fall between the 58 and 79 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

set for an investor having access to the Fama-French three-factor (six-factor) model. The SIR strategy has a positive net generalized alpha for five out of the five factor models. In these cases SIR ranks between the 78 and 90 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does SIR add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of SIR with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price SIR or at least to weaken the power SIR has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of SIR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{SIR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{SIR}SIR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{SIR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on SIR. Stocks are finally grouped into five SIR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted SIR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on SIR and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the SIR signal in these Fama-MacBeth regressions exceed 1.09, with the minimum t-statistic occurring when controlling for Momentum and LT Reversal. Controlling for all six closely related anomalies, the t-statistic on SIR is -0.61.

Similarly, Table 5 reports results from spanning tests that regress returns to the SIR strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the SIR strategy earns alphas that range from 19-27bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.55, which is achieved when controlling for Momentum and LT Reversal. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the SIR trading strategy achieves an alpha of 20bps/month with a t-statistic of 2.32.

7 Does SIR add relative to the whole zoo?

Finally, we can ask how much adding SIR to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the SIR signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes SIR grows to \$2963.61.

8 Conclusion

This study provides robust evidence that Share Issuance Restraint (SIR) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that SIR delivers substantial risk-adjusted returns that persist even after controlling for well-established factors and closely related anomalies. The value-weighted long/short strategy based on SIR generates an annualized net Sharpe ratio of 0.45, which is economically significant and comparable to many established factors in the literature. More importantly, the signal maintains statisti-

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which SIR is available.

cal and economic significance with a monthly alpha of 25 basis points (t-statistic = 3.27) relative to the Fama-French five-factor model plus momentum, indicating that SIR captures unique information about expected returns not subsumed by existing factors.

The practical implications of our findings are noteworthy for both academics and practitioners. For investors, the relatively modest degradation from gross to net returns (from 0.51 to 0.45 Sharpe ratio) suggests that the strategy remains viable after accounting for realistic transaction costs. Furthermore, the persistence of a 20 basis points monthly alpha even after controlling for six closely related strategies—including various share issuance measures, growth metrics, and analyst-based signals—underscores that SIR contains distinct information content beyond simple share dilution effects or growth characteristics.

Despite these encouraging results, several limitations warrant consideration. First, while our analysis accounts for transaction costs, implementation challenges such as market impact and capacity constraints in real-world trading environments may further erode returns. Second, the sample period and market conditions examined may not fully capture the strategy’s performance across different market regimes or during periods of significant market stress. Third, the economic mechanism underlying SIR’s predictive power requires further theoretical development to fully understand why firms’ restraint in issuing shares predicts future returns beyond what is captured by existing issuance-based measures.

Future research should explore several promising avenues. First, investigating the time-varying nature of the SIR premium and its relationship to macroeconomic conditions could provide insights into when the strategy performs best. Second, examining the international evidence for SIR would help establish whether this is a global phenomenon or specific to certain markets. Third, decomposing the sources of SIR’s predictive power—whether it stems from behavioral biases, limits to arbitrage,

or risk-based explanations—would enhance our theoretical understanding. Finally, exploring potential enhancements to the SIR signal through machine learning techniques or combination with other predictors could yield strategies with even stronger performance characteristics. Overall, our findings suggest that Share Issuance Restriction represents a valuable addition to the factor zoo, offering both academic insights and practical investment applications.

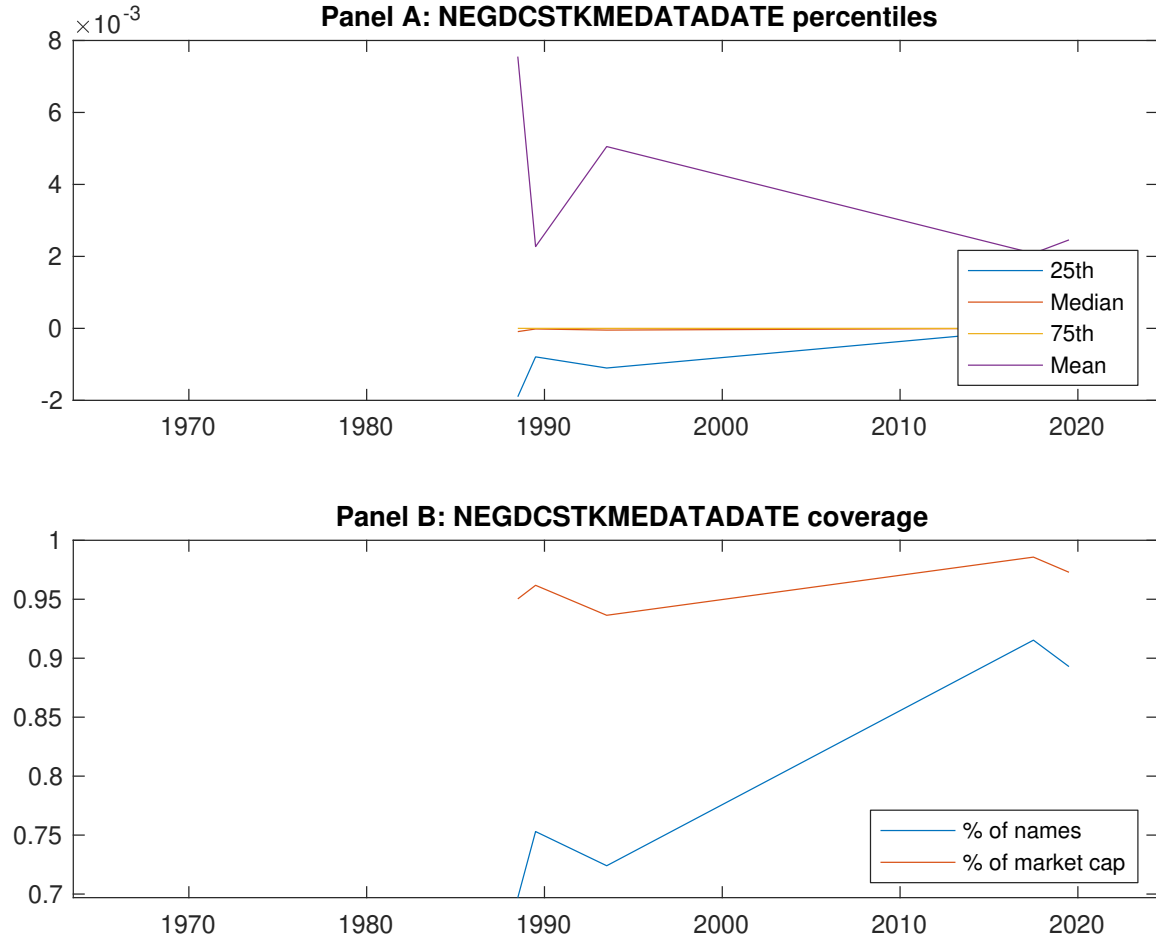


Figure 1: Times series of SIR percentiles and coverage. This figure plots descriptive statistics for SIR. Panel A shows cross-sectional percentiles of SIR over the sample. Panel B plots the monthly coverage of SIR relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on SIR. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on SIR-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.45 [2.77]	0.53 [2.87]	0.61 [3.33]	0.63 [3.92]	0.75 [4.69]	0.30 [4.01]
α_{CAPM}	-0.09 [-1.63]	-0.11 [-2.37]	-0.02 [-0.42]	0.08 [1.78]	0.21 [4.42]	0.30 [3.93]
α_{FF3}	-0.14 [-2.66]	-0.10 [-2.15]	0.02 [0.48]	0.06 [1.28]	0.16 [3.64]	0.31 [4.00]
α_{FF4}	-0.13 [-2.38]	-0.08 [-1.80]	0.05 [1.14]	0.03 [0.57]	0.15 [3.40]	0.29 [3.66]
α_{FF5}	-0.21 [-3.83]	-0.04 [-0.84]	0.07 [1.43]	-0.02 [-0.39]	0.09 [2.05]	0.30 [3.84]
α_{FF6}	-0.19 [-3.51]	-0.03 [-0.68]	0.09 [1.93]	-0.04 [-0.84]	0.09 [2.03]	0.28 [3.60]
Panel B: Fama and French (2018) 6-factor model loadings for SIR-sorted portfolios						
β_{MKT}	0.96 [72.78]	1.04 [94.53]	1.03 [90.27]	0.99 [92.56]	0.98 [91.50]	0.02 [1.20]
β_{SMB}	0.01 [0.44]	0.03 [1.64]	-0.00 [-0.25]	-0.06 [-3.84]	-0.01 [-0.95]	-0.02 [-0.84]
β_{HML}	0.14 [5.39]	0.02 [0.71]	-0.12 [-5.46]	0.04 [1.70]	0.05 [2.33]	-0.09 [-2.44]
β_{RMW}	0.18 [7.14]	-0.10 [-4.50]	-0.10 [-4.69]	0.12 [5.66]	0.10 [4.98]	-0.08 [-2.15]
β_{CMA}	0.01 [0.22]	-0.13 [-4.07]	-0.03 [-0.78]	0.15 [4.95]	0.18 [6.06]	0.18 [3.28]
β_{UMD}	-0.02 [-1.61]	-0.01 [-0.91]	-0.03 [-3.04]	0.03 [2.71]	-0.00 [-0.05]	0.02 [1.10]
Panel C: Average number of firms (n) and market capitalization (me)						
n	694	693	617	684	727	
me (\$10 ⁶)	1465	1493	2211	2378	2579	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the SIR strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.30 [4.01]	0.30 [3.93]	0.31 [4.00]	0.29 [3.66]	0.30 [3.84]	0.28 [3.60]
Quintile	NYSE	EW	0.44 [8.02]	0.49 [8.92]	0.44 [8.28]	0.39 [7.26]	0.35 [6.58]	0.31 [5.86]
Quintile	Name	VW	0.30 [3.97]	0.30 [3.88]	0.30 [3.88]	0.29 [3.64]	0.30 [3.88]	0.29 [3.71]
Quintile	Cap	VW	0.25 [3.41]	0.25 [3.42]	0.27 [3.56]	0.23 [3.03]	0.23 [3.02]	0.20 [2.67]
Decile	NYSE	VW	0.32 [3.85]	0.30 [3.61]	0.29 [3.44]	0.25 [2.92]	0.24 [2.83]	0.21 [2.48]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.26 [3.49]	0.26 [3.43]	0.26 [3.44]	0.25 [3.27]	0.26 [3.39]	0.25 [3.27]
Quintile	NYSE	EW	0.25 [3.94]	0.31 [4.93]	0.26 [4.29]	0.23 [3.87]	0.16 [2.74]	0.14 [2.48]
Quintile	Name	VW	0.26 [3.46]	0.26 [3.41]	0.26 [3.35]	0.25 [3.24]	0.26 [3.44]	0.26 [3.36]
Quintile	Cap	VW	0.21 [2.92]	0.22 [2.95]	0.22 [3.02]	0.20 [2.74]	0.20 [2.68]	0.18 [2.50]
Decile	NYSE	VW	0.27 [3.32]	0.25 [3.00]	0.24 [2.85]	0.22 [2.59]	0.20 [2.37]	0.18 [2.20]

Table 3: Conditional sort on size and SIR

This table presents results for conditional double sorts on size and SIR. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on SIR. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high SIR and short stocks with low SIR. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	SIR Quintiles					SIR Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.54 [2.24]	0.72 [2.79]	0.75 [2.85]	0.91 [3.82]	1.00 [4.44]	0.48 [7.02]	0.51 [7.49]	0.49 [7.21]	0.44 [6.40]	0.42 [6.09]	0.38 [5.50]
	(2)	0.66 [2.98]	0.67 [2.87]	0.85 [3.54]	0.90 [3.97]	0.94 [4.36]	0.27 [3.42]	0.30 [3.75]	0.26 [3.21]	0.23 [2.84]	0.22 [2.69]	0.20 [2.45]
	(3)	0.59 [3.01]	0.67 [3.09]	0.69 [3.14]	0.83 [4.04]	0.94 [4.87]	0.36 [4.83]	0.36 [4.81]	0.34 [4.55]	0.32 [4.19]	0.32 [4.13]	0.30 [3.88]
	(4)	0.51 [2.72]	0.68 [3.43]	0.78 [3.81]	0.71 [3.74]	0.81 [4.48]	0.30 [4.34]	0.33 [4.63]	0.29 [4.20]	0.30 [4.21]	0.16 [2.33]	0.18 [2.58]
	(5)	0.50 [3.09]	0.48 [2.68]	0.52 [2.95]	0.50 [3.06]	0.70 [4.48]	0.21 [2.36]	0.20 [2.31]	0.22 [2.44]	0.18 [2.02]	0.20 [2.20]	0.17 [1.90]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	SIR Quintiles					SIR Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	379	379	380	378	372	31	32	42	30	29	
	(2)	109	107	108	108	106	57	56	58	57	55	
	(3)	78	78	77	77	78	98	94	98	100	100	
	(4)	66	65	66	66	66	204	204	216	212	217	
(5)	60	60	60	60	60	1431	1407	1661	1742	1900		

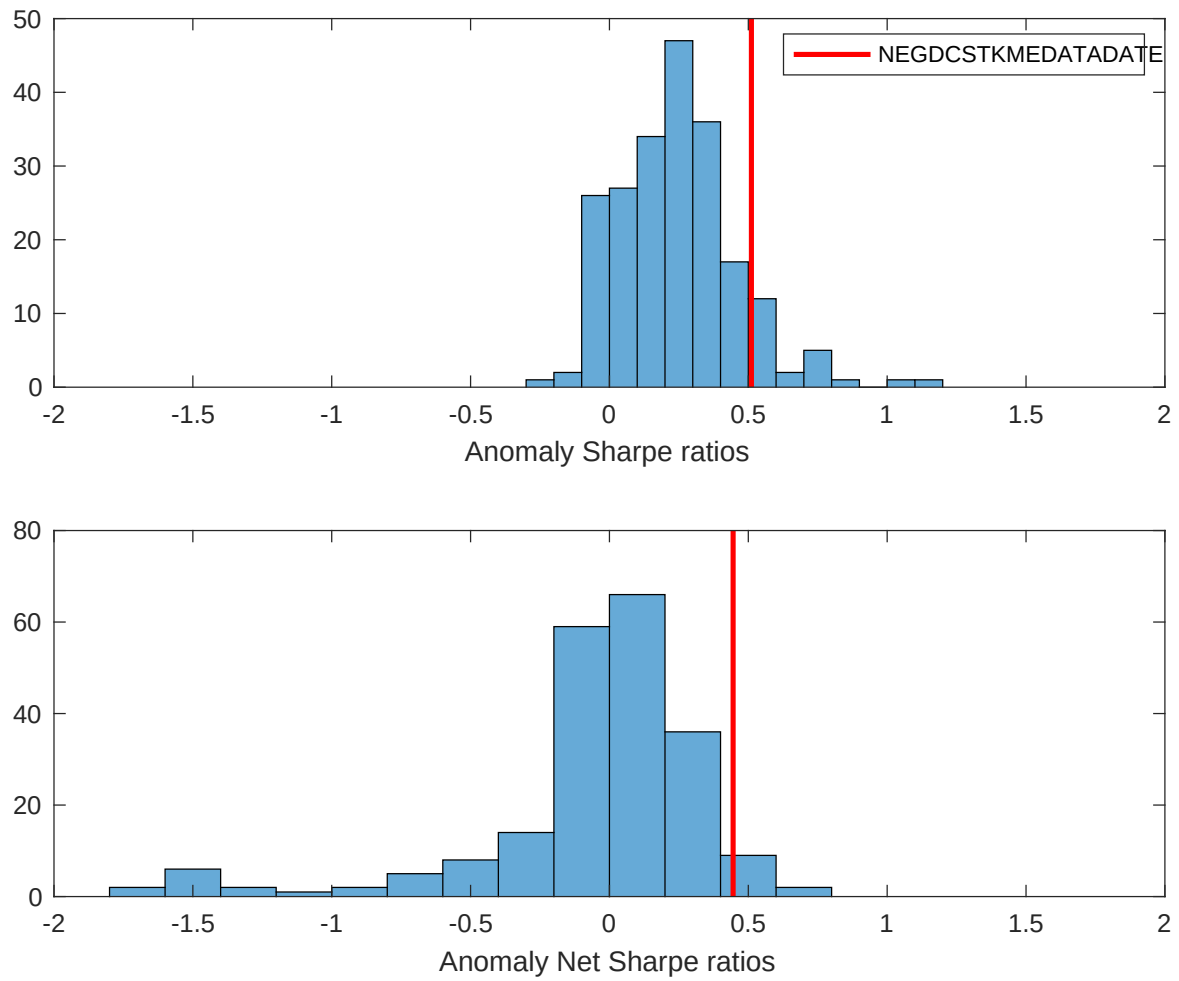


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the SIR with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

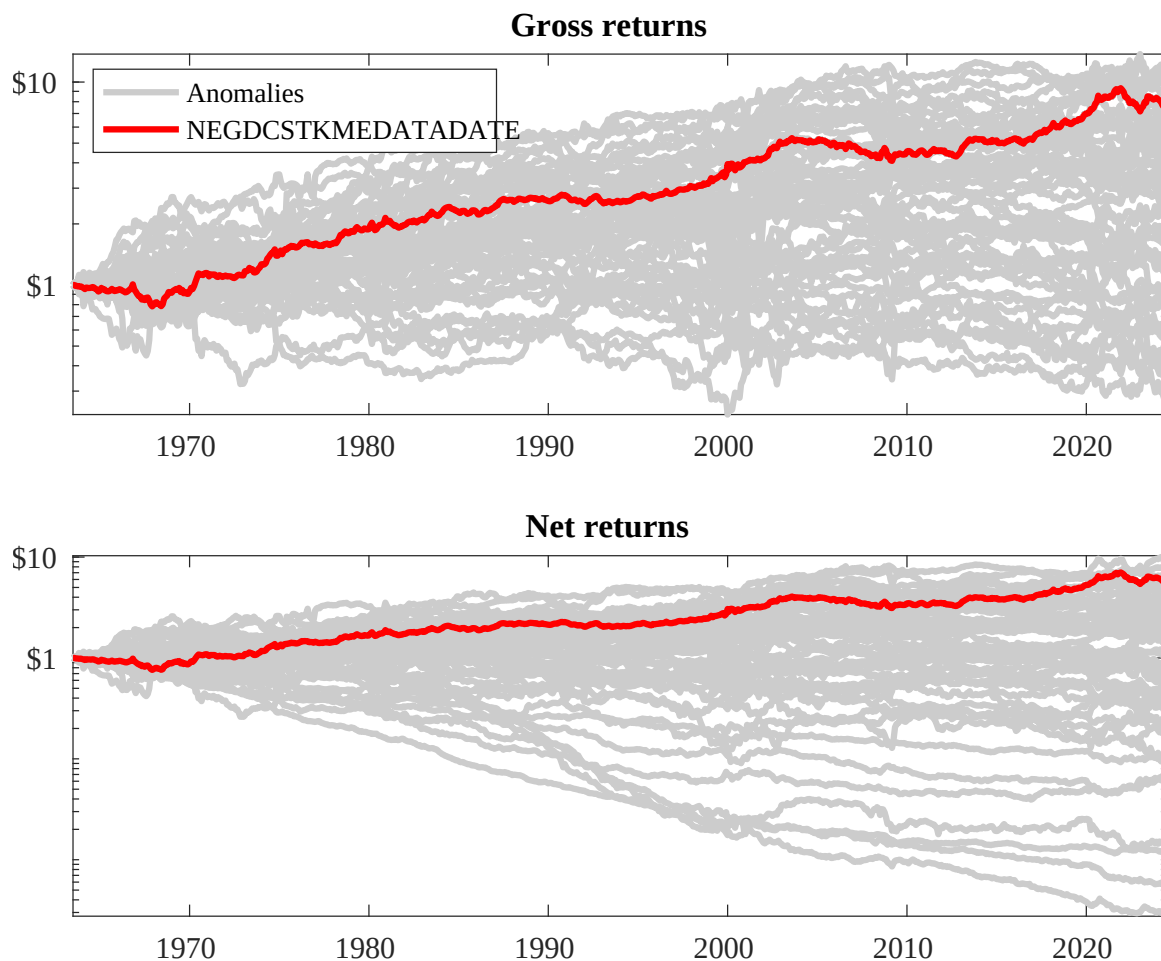
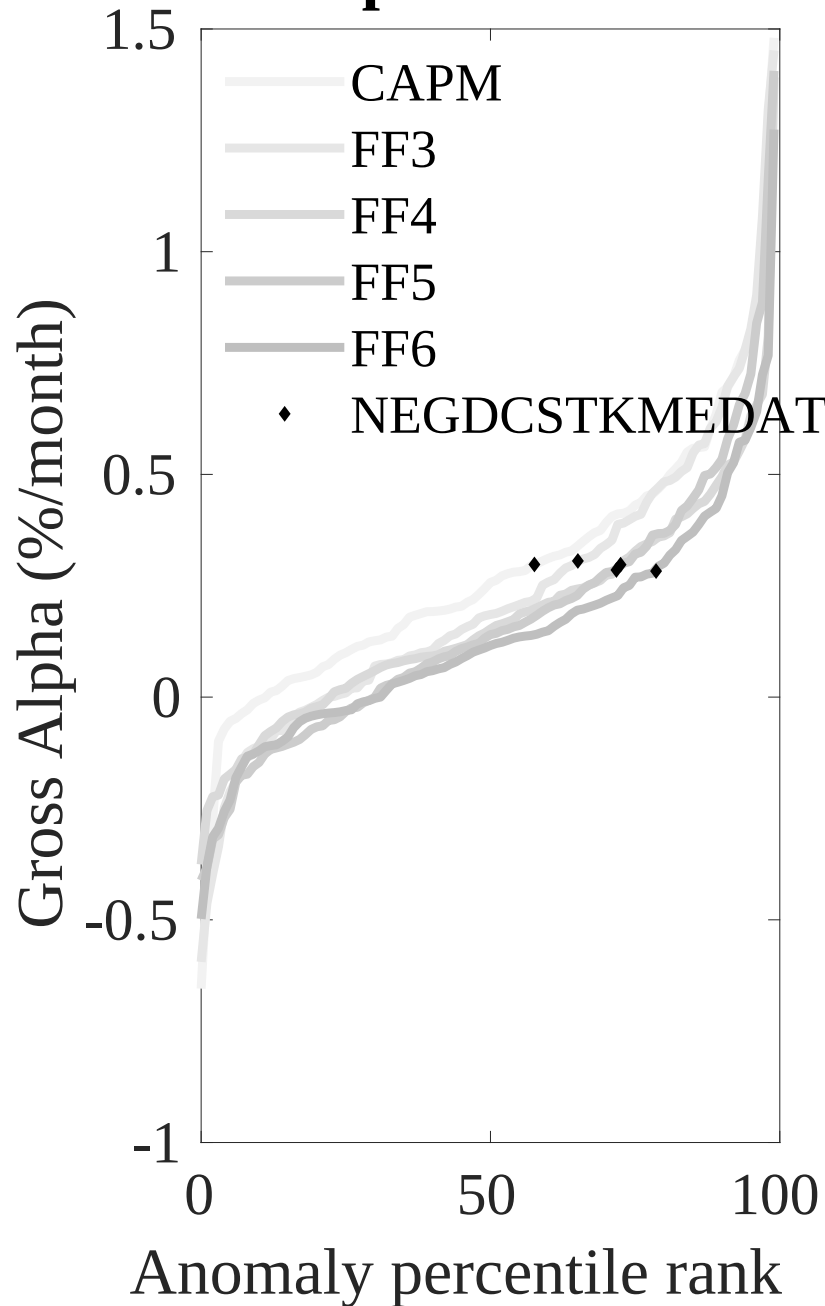


Figure 3: Dollar invested.

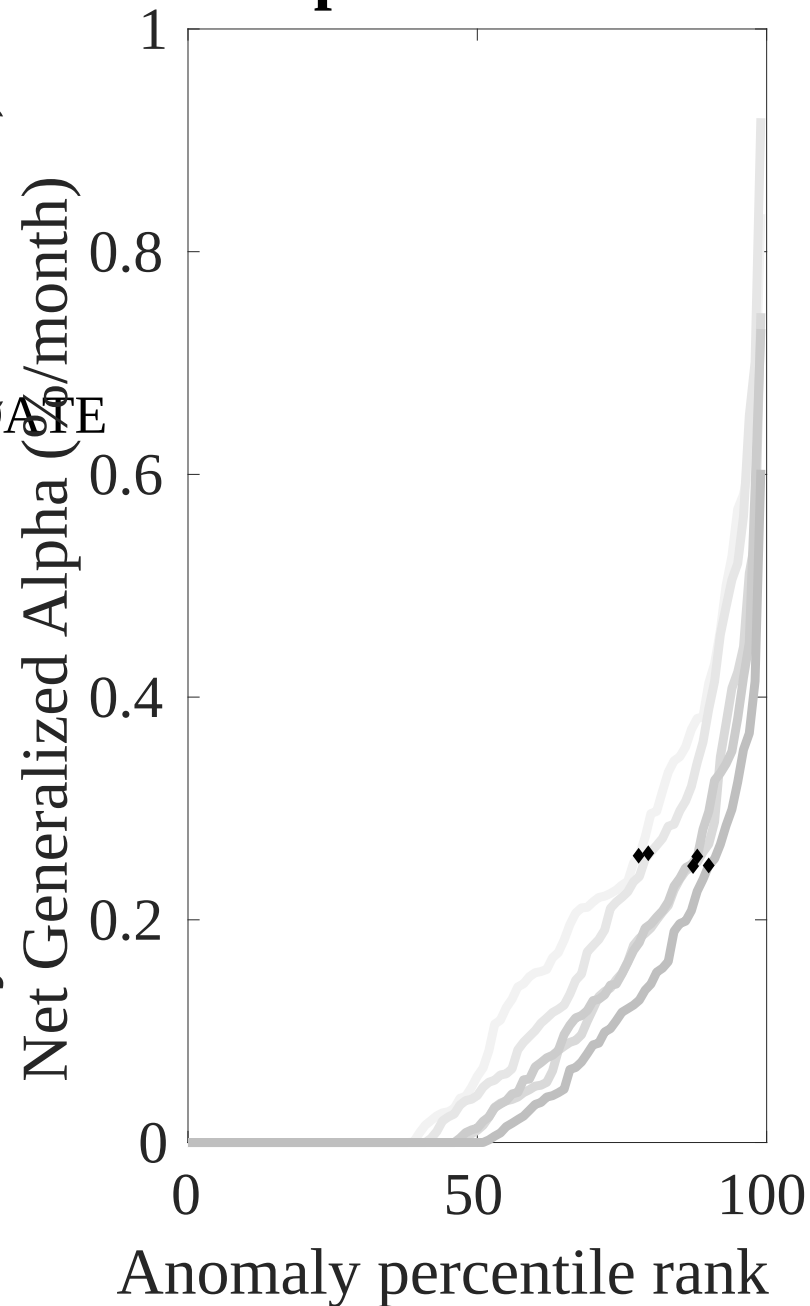
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the SIR trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (2016)



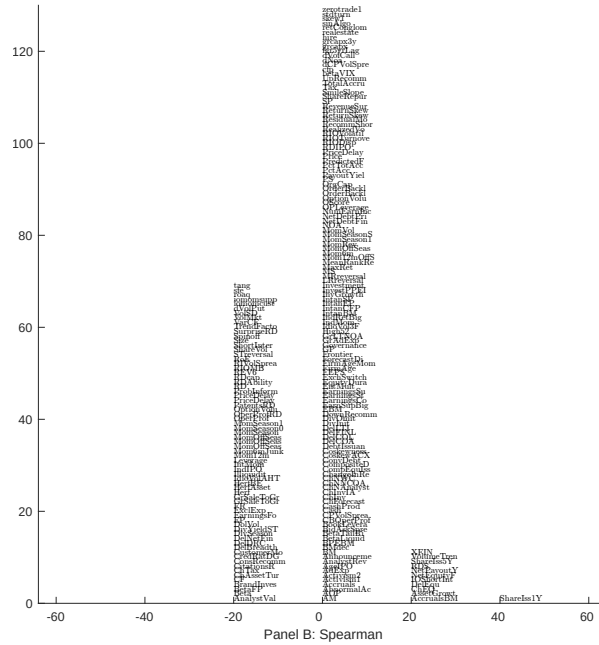
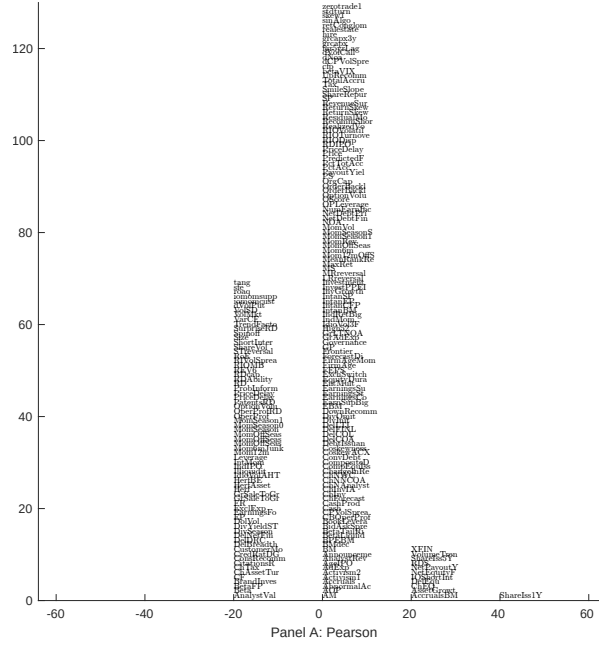


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 210 filtered anomaly signals with SIR. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

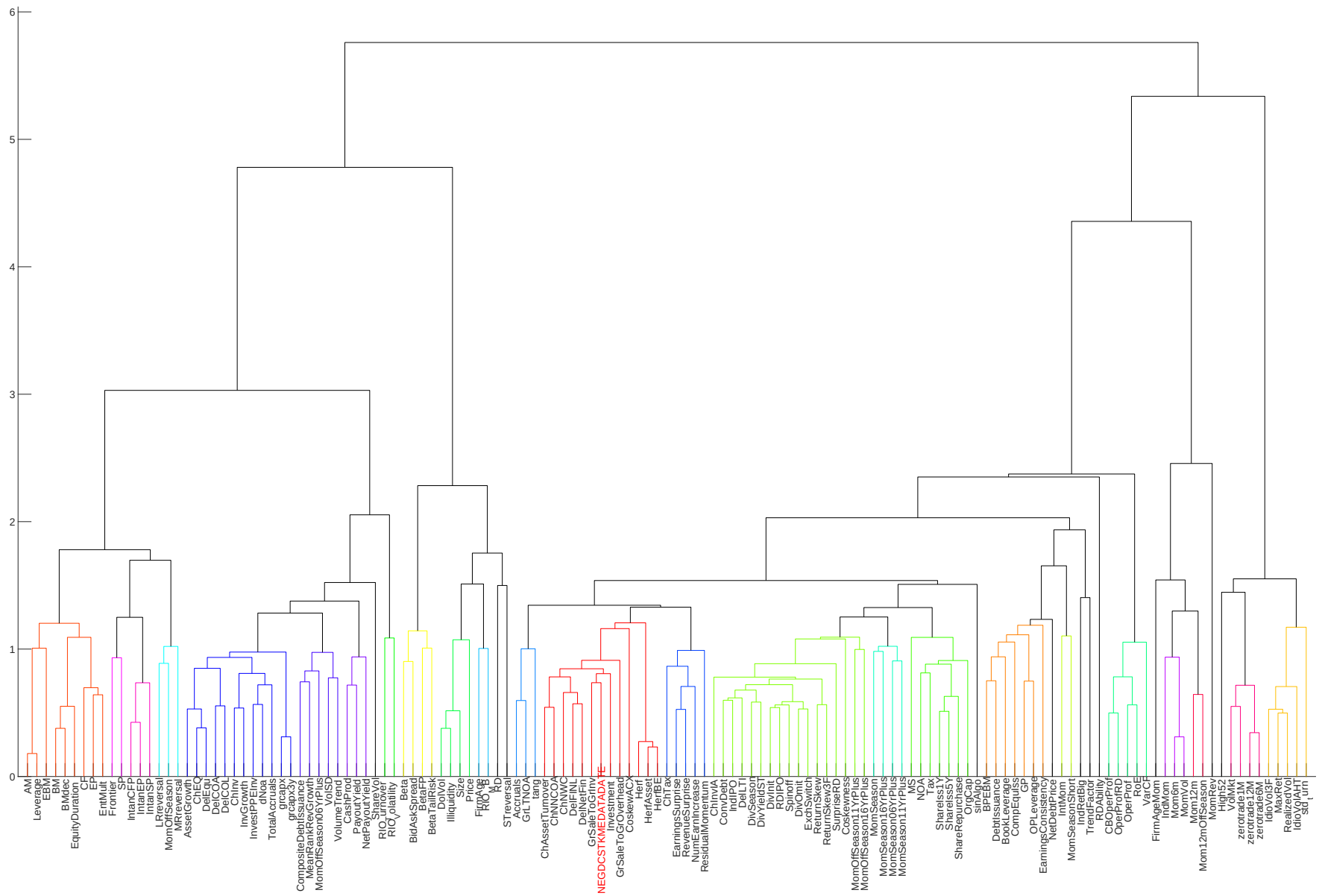


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

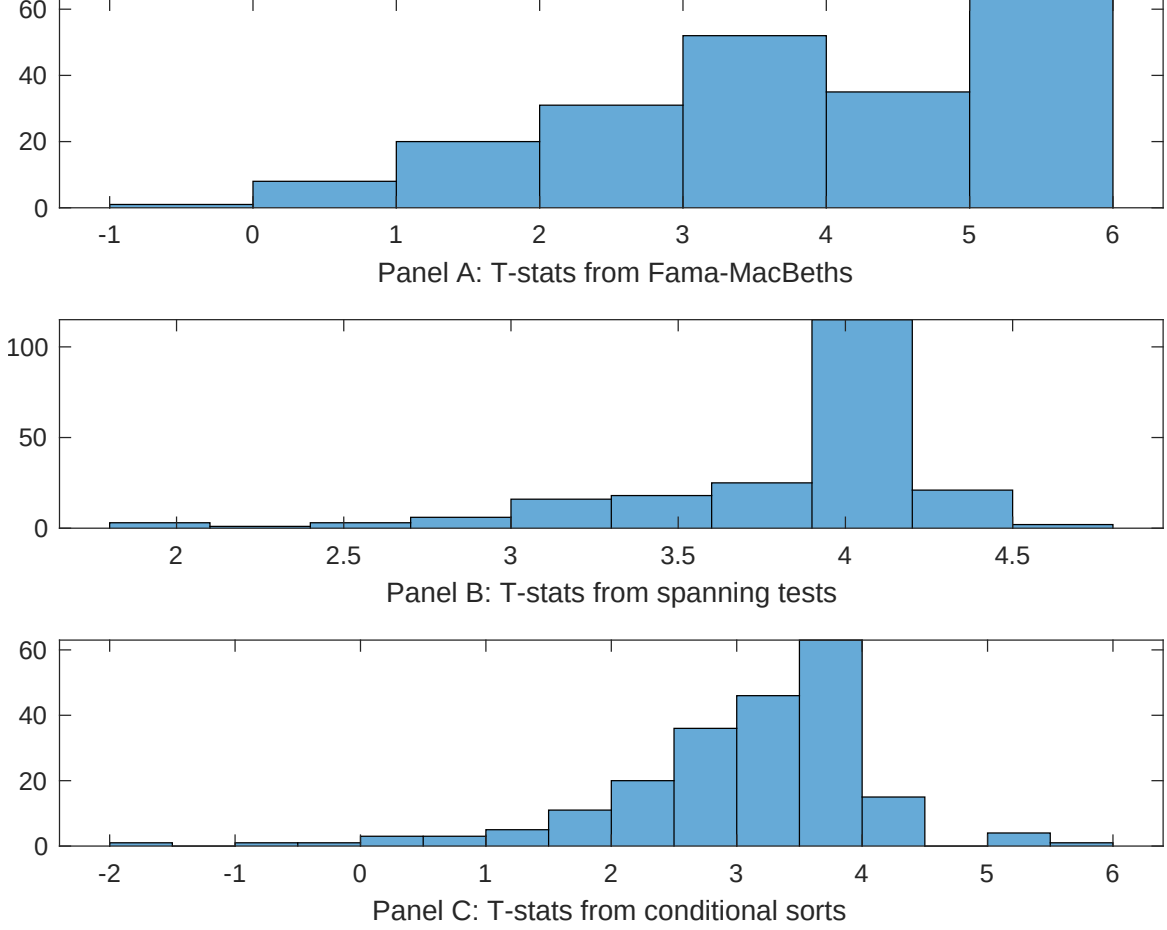


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of SIR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{SIR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{SIR}SIR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{SIR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on SIR. Stocks are finally grouped into five SIR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted SIR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on SIR. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{SIR}SIR_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Share issuance (5 year), Growth in book equity, Share issuance (1 year), Momentum and LT Reversal, Long-term EPS forecast, Analyst Value. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [6.41]	0.17 [6.99]	0.13 [6.01]	0.39 [1.14]	0.14 [6.58]	0.11 [4.41]	0.16 [0.56]
SIR	0.31 [3.47]	0.39 [3.16]	0.34 [3.52]	0.50 [1.09]	0.39 [2.04]	0.50 [3.91]	-0.28 [-0.61]
Anomaly 1	0.27 [4.86]						0.86 [2.24]
Anomaly 2		0.41 [3.48]					0.14 [1.14]
Anomaly 3			0.13 [5.85]				-0.25 [-0.71]
Anomaly 4				0.11 [4.25]			-0.18 [-0.09]
Anomaly 5					0.10 [1.26]		-0.14 [-1.36]
Anomaly 6						0.71 [0.67]	0.73 [0.44]
# months	715	720	715	648	504	576	14
$\bar{R}^2(\%)$	0	0	0	2	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the SIR trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{SIR} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Share issuance (5 year), Growth in book equity, Share issuance (1 year), Momentum and LT Reversal, Long-term EPS forecast, Analyst Value. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.19 [2.55]	0.27 [3.53]	0.26 [3.42]	0.25 [3.28]	0.27 [2.93]	0.25 [2.95]	0.20 [2.32]
Anomaly 1	32.35 [8.07]						12.10 [2.10]
Anomaly 2		25.53 [6.18]					22.50 [4.11]
Anomaly 3			26.21 [6.82]				15.58 [2.75]
Anomaly 4				4.49 [4.26]			3.49 [3.26]
Anomaly 5					-12.62 [-4.07]		-17.57 [-5.83]
Anomaly 6						-9.45 [-3.32]	-14.50 [-4.75]
mkt	3.74 [2.09]	3.28 [1.80]	2.95 [1.63]	2.45 [1.33]	-2.49 [-1.04]	5.49 [2.73]	-1.08 [-0.48]
smb	1.16 [0.44]	-1.97 [-0.75]	-0.01 [-0.00]	-2.82 [-1.05]	-1.47 [-0.43]	4.02 [1.30]	-2.83 [-0.88]
hml	-5.07 [-1.46]	-10.62 [-2.99]	-8.27 [-2.36]	-8.65 [-2.42]	-3.84 [-0.88]	-6.46 [-1.54]	6.89 [1.59]
rmw	-19.94 [-5.28]	-6.81 [-1.92]	-22.52 [-5.49]	-6.01 [-1.66]	2.37 [0.56]	5.70 [1.40]	-4.40 [-0.93]
cma	1.48 [0.27]	-8.09 [-1.24]	0.82 [0.14]	14.88 [2.83]	12.91 [1.98]	5.46 [0.95]	-22.10 [-3.03]
umd	0.13 [0.07]	1.80 [1.00]	1.06 [0.59]	-2.11 [-1.01]	-0.44 [-0.21]	0.91 [0.46]	-6.03 [-2.73]
# months	728	732	728	728	504	576	500
$\bar{R}^2(\%)$	10	7	8	4	5	5	21

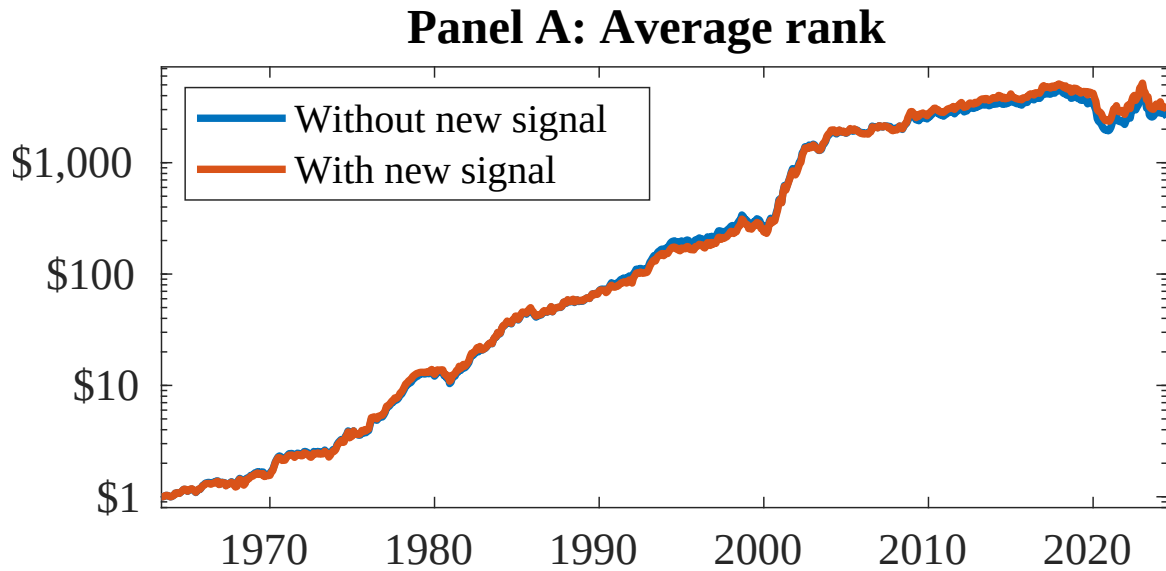


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as SIR. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

References

- Bansal, R. and Yaron, A. (2004). Risks for the long run: A potential resolution of asset pricing puzzles. *Journal of Finance*, 59(4):1481–1509.
- Barro, R. J. (2006). Rare disasters and asset markets in the twentieth century. *Quarterly Journal of Economics*, 121(3):823–866.
- Breeden, D. T. and Litzenberger, R. H. (1978). Prices of state-contingent claims implicit in option prices. *Journal of Business*, 51(4):621–651.
- Campbell, J. Y. and Cochrane, J. H. (1999). By force of habit: A consumption-based explanation of aggregate stock market behavior. *Journal of Political Economy*, 107(2):205–251.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- Cochrane, J. H. (2005). *Asset Pricing*. Princeton University Press, revised edition.
- Cochrane, J. H., Piazzesi, M., and Schneider, M. (2008). Decomposing the yield curve. *Working Paper*.
- Constantinides, G. M. (1990). Habit formation: A resolution of the equity premium puzzle. *Journal of Political Economy*, 98(3):519–543.
- Daniel, K. and Titman, S. (2006). Market reactions to tangible and intangible information. *Journal of Finance*, 61(4):1605–1643.

- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, *Forthcoming*.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.
- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.
- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.
- Gabaix, X. (2012). Variable rare disasters: An exactly solved framework for ten puzzles in macro-finance. *Quarterly Journal of Economics*, 127(2):645–700.
- Harvey, C. R., Liu, Y., and Zhu, H. (2016). ... and the cross-section of expected returns. *Review of Financial Studies*, 29(1):5–68.
- Hou, K., Xue, C., and Zhang, L. (2020). Replicating anomalies. *Review of Financial Studies*, 33(5):2019–2133.
- McLean, R. D. and Pontiff, J. (2016). Does academic research destroy stock return predictability? *Journal of Finance*, 71(1):5–32.
- Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.
- Pontiff, J. and Woodgate, A. (2008). Share issuance and cross-sectional returns. *Journal of Finance*, 63(2):921–945.
- Rietz, T. A. (1988). The equity risk premium: A solution. *Journal of Monetary Economics*, 22(1):117–131.